OSI vs TCP/IP model: table of layers, protocols and one attack per layer.

Network models explain how data travels between devices over a network. They divide communication into layers, with each layer performing specific functions.

The two important networking models are:

- **OSI Model**: A conceptual model consisting of seven layers that helps explain network communication and troubleshoot problems.
- **TCP/IP Model**: A practical networking model consisting of four layers that describes the protocols used for communication over the internet.

OSI Model - Seven Layers

The OSI (Open Systems Interconnection) model divides network communication into seven layers.

Layer	Name	Main function	Example protocols or technologies
7	Application	Provides network services to applications.	HTTP, DNS, SMTP, FTP
6	Presentation	Handles data representation, encoding, and encryption-related functions.	Character encoding, data formats, TLS-related functions
5	Session	Establishes, manages, and terminates communication sessions.	RPC, session-management mechanisms
4	Transport	Provides end-to-end data transport and, with TCP, reliable delivery.	TCP, UDP
3	Network	Handles logical addressing and routing between networks.	IPv4, IPv6, ICMP
2	Data Link	Transfers frames across a local network and uses MAC addressing.	Ethernet, Wi-Fi, ARP
1	Physical	Transmits raw bits through physical or radio signals.	Cables, fiber optics, radio signals

Security Attack or Risk Per OSI Layer

OSI layer	Example attack or risk	Explanation
Application	SQL injection	A vulnerable application may allow malicious input to interfere with database queries.
Presentation	Weak encryption or downgrade attack	Inadequate cryptographic configuration may expose communications.
Session	Session hijacking	An attacker who obtains a valid session token may impersonate a user.
Transport	TCP SYN flood	Excessive connection requests can consume server resources.
Network	IP spoofing	An attacker falsifies a packet's source IP address.
Data Link	ARP spoofing	An attacker on a local network sends forged ARP messages to misdirect traffic.
Physical	Cable tampering	Unauthorized physical access can disrupt or intercept network connections.

TCP/IP Model - Four Layers

The TCP/IP model describes the practical architecture used for internet communication.

TCP/IP layer	Main function	Example protocols
Application	Provides services used by applications.	HTTP, HTTPS, DNS, SMTP, SSH
Transport	Provides communication between applications using port numbers.	TCP, UDP
Internet	Handles IP addressing and routing between networks.	IPv4, IPv6, ICMP
Network Access (Link)	Handles communication over the local network medium.	Ethernet, Wi-Fi

OSI vs TCP/IP Model

Feature	OSI Model	TCP/IP Model
Number of layers	Seven	Four in the common model
Purpose	Reference model for understanding	Practical model for internet

	network communication.	communication.
Development	Developed through ISO standardization.	Developed through the Internet protocol suite.
Layer separation	Separates application, presentation, and session functions.	Groups these functions into the application layer.
Usage	Learning, troubleshooting, and explaining network functions.	Designing, implementing, and understanding real-world networking.

Mapping the Models

OSI layers	**Corresponding TCP/IP layer**
Application, Presentation, Session (7–5)	Application
Transport (4)	Transport
Network (3)	Internet
Data Link, Physical (2–1)	Network Access / Link

The mapping is approximate because the models organize networking functions differently.